

Varia (in 5')

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2026/09/09

- CDB
 - ▷ detector config/calibration files
 - ▷ v7.X global tags
- Prompt
 - ▷ a bit terse
- RDB
 - ▷ changes for 2026
 - ▷ JSON backend
- Parts DB

CDB

- CDB - catch-all term for all non-(O,parts)DB in mu3e

- **Notabene**

- ▷ **backends**

- node + REST (mongodb)
 - files

- ▷ **access**

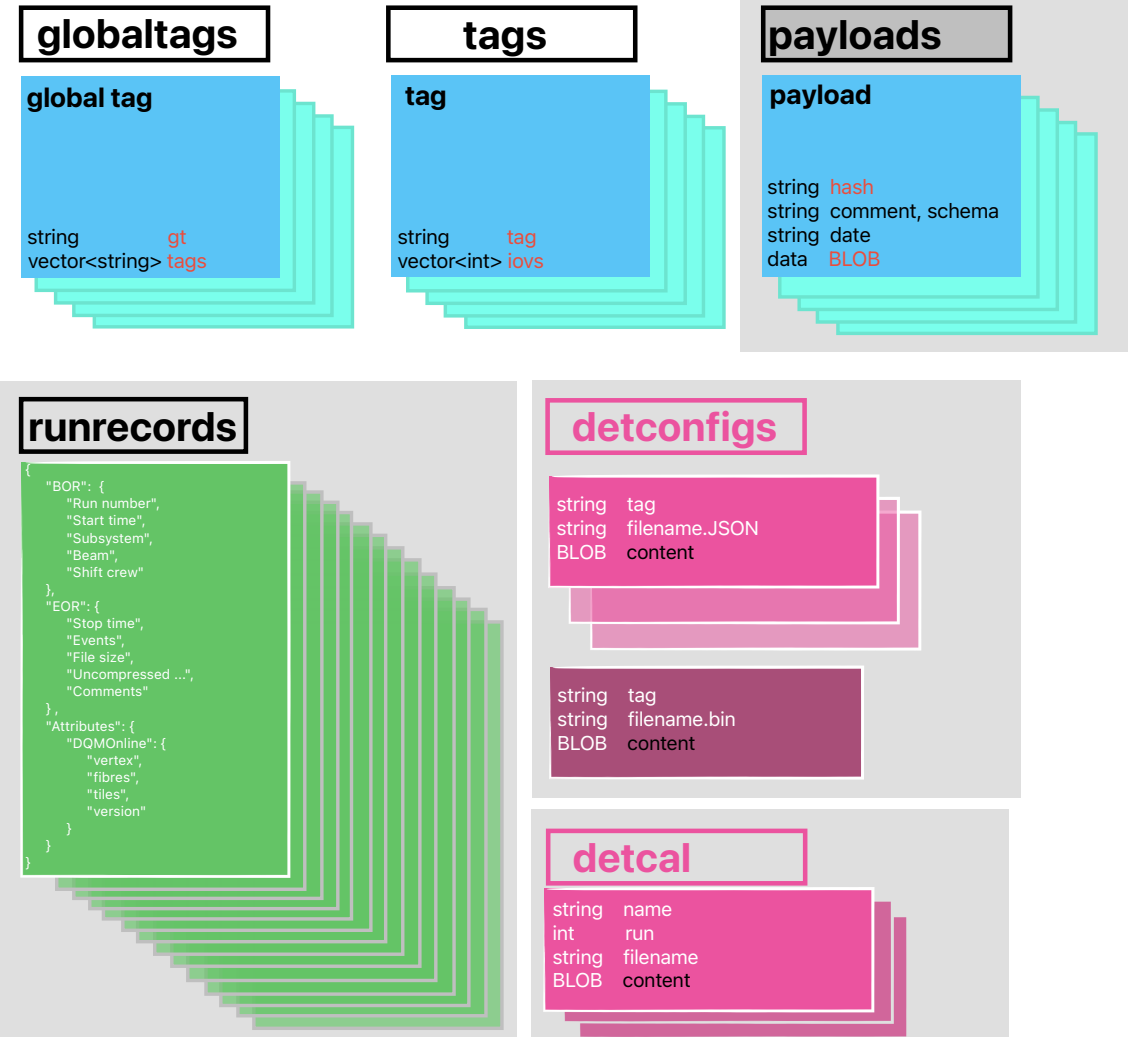
- REST (http)
 - file reading
 - mu3eUtil/cdb

- ▷ **tools**

- browser (REST (mongodb) and files)
 - (many) CLI programs

- ▷ **documentation**

- [Mu3e-Note-0133](#)
 - ([Mu3e-Note-0133](#)) stuck . . .
 - [CDB slides](#)



Detector configs/calibrations

usual PSI firewall restrictions apply
(but **detector** at PSI, so not a real issue)

- People asked for versioned storage of many/large files in CDB
 - ▷ detector configs: e.g. pixel mask files
 - ▷ detector calibration files: no example exists yet

- **det configs** (multi file up/download also possible)
 - ▷ files are stored in CDB, with a "tag" ("label"/"setup name")

```
curl -X POST -F "tag=tdac_bu_06_10" -F "file=@/home/mu3e/junk/tdac_files_bu_06_10/mask_chip_0.bin" \
  http://mu3edb0/detconfigs/upload
curl -fSL --output tdac_files_bu_06_10.zip "http://mu3edb0/detconfigs/downloadTag?tag=tdac_files_bu_06_10"
```

- ▷ see [Mu3e-Note-0133](#) for more examples and translation to python **requests**

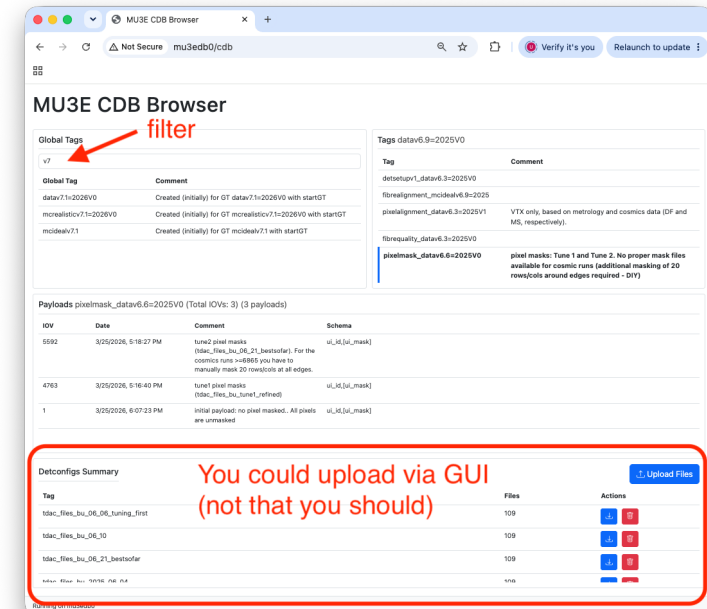
- **det calibrations**
 - ▷ calibration files are stored in CDB, with a "(calibration) name" and "run"

```
curl -X POST -F "name=pp0" -F "run=5700" -F "file=@/home/mu3e/junk/detector0.json" http://mu3edb0/detcal/upload
curl -OJ "http://mu3edb0/detcal/pp0/6000"
```

- ▷ notabene
 - different use case than det configs! **Talk to me if this is unclear**
 - "run" $\approx$ interval of validity and corresponding access (as in CDB proper)
 - much more possible, see [Mu3e-Note-0133](#) for more and translation to **requests**

Global tags for 2026 data

- Mu3e v7.X for 2026 datataking and MC production (I guess?)
 - ▷ 2026 detector in alignment (= geometry) tags
 - VTX + OPC L3 (naming??)
 - complete fibres
 - DS tiles
- Starting point global tags (GTs) prepared
 - ▷ **mcidealv7.1** - with complete(!) detector
 - ▷ **mcrealisticv7.1=2026V0** - for realistic MC
 - ▷ **datav7.1=2026V0** - initial payload for run 1
- **Notabene**
 - ▷ mcrealistic currently mirrors mcideal
 - **should replace "ideal" with "correct" alignment (=geometry)**
at least for me unclear whether "data" alignment is the "correct" one (not my job to figure this out or to produce the payload :-)
 - maybe also for other "calibrations" (e.g. (pixel)timecalibration)
 - ▷ **prompt** will supply most of the "quality" payloads for **realistic detector** (prompt I consider to be my job)



Prompt

- Prompt **what?**
 - ▷ DQM - full statistics, for reference uploaded to RDB
 - ▷ calibration - identify deficient detector elements, etc.
 - ▷ reconstruction - possibly w/ knowledge of deficient detector elements
 - within a "reasonable" time (keep up with data taking)
- Different from "reconstruction" or "re-processing"
 - ▷ e.g. address DAQ issues or apply ultimate alignment
- For details see [DQM meeting presentation of 2026/04/21](#)
 - ▷ re-implemented (modular), tested, and mostly running on merlin-6
 - ▷ Complete documentation at [prod/prompt0](#)
 - ▷ runs
 - midasmeta, skipsmallruns, "minalyzer", minalyzer_pdf
 - pixelqualitylm, other quality determinations
 - updates to RDB and CDB
 - mu3esort, mu3etrirec
 - still missing: a few "calibrations"
 - **waiting for data**

RDB - Run Database

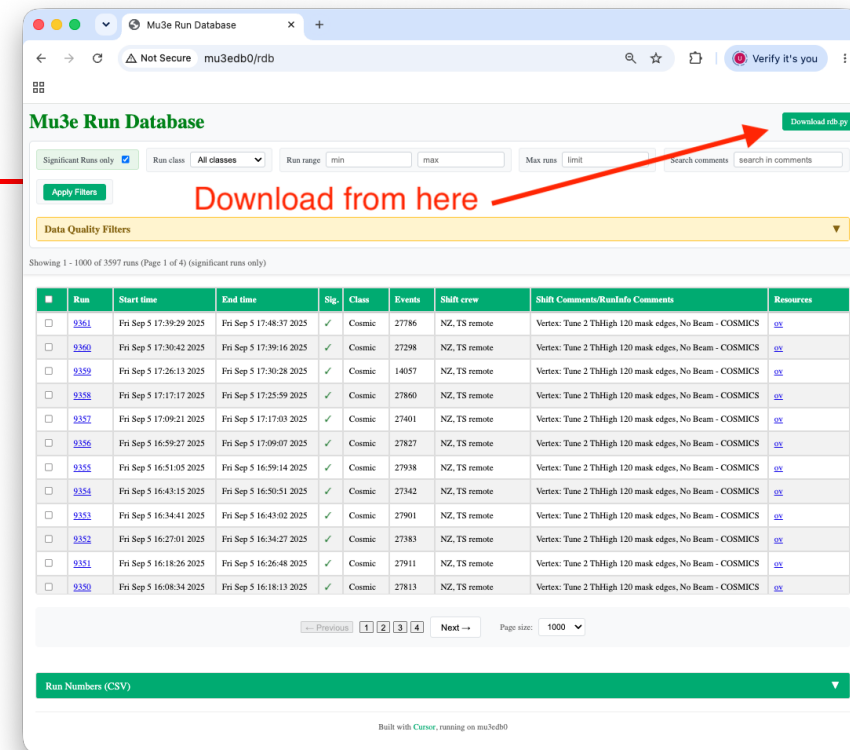
- The RDB will receive a few updates
 - ▷ year selection
 - ▷ revision of contents
 - ▷ soon (before datataking starts)
- [rdb.py](#) for better local access
 - ▷ e.g.

```
rdb.py --dir ~/data/mu3e/cdb --sync
rdb.py --dir ~/data/mu3e/cdb --sync --all -f 4000 -l 4100
rdb.py --dir ~/data/mu3e/cdb --rebuild
rdb.py --dir ~/data/mu3e/cdb --significant
rdb.py --dir ~/data/mu3e/cdb --class cosmic --significant --r 2000-3000
rdb.py --dir ~/data/mu3e/cdb -f 4000 -l 5000 --min-events 10000
rdb.py --dir ~/data/mu3e/cdb --class beam --format json
rdb.py --dir ~/data/mu3e/cdb --significant --format line --summary
rdb.py --dir ~/data/mu3e/cdb -r 226 --format dump
rdb.py --update
```

- ▷ Full documentation in help message or in Sec 9.3 of [Mu3e-Note-0133](#)
- ▷ just started - please provide feedback on issues/additional features

⇒ **The RDB is only as useful as the information put in by shifters!**

- ▷ should we discuss "eras"?



Parts DB

- Not much activity in the past 2 years
 - ▷ minor user support, once every while
 - ▷ information sources
 - mattermost [mu3epartsdb](#)
 - mattermost [database-helpdesk](#)
 - post to mattermost for feature requests, now that bitbucket dropped issues
- New features for **product document upload**
 - ▷ drag-n-drop for files
 - ▷ delete button
- "mirror"/"backup" possible
 - ▷ mu3edb0.psi.ch
 - ▷ migration within "a few days" (maybe ssh tunneling necessary for outside access)

Documents

Type	Filename	Size	Comment	
drawing	2021-Paracetamol-s41574-021-00553-7.pdf	1.5 MB	asdasd	
picture	2026_08_10_1415.JPG	11.2 MB	qwerty	

Upload new document

Document type:

Comment:

File:

Drop a file here, or choose a file

No file chosen

No file selected

Conclusions

- Since ≥ 1 year you can store **detector config files** in CDB
 - ▷ unused. **Do we care about which mask version(s) will be used in 2026?**
- Since ≥ 1 month you can store **detector calibration files** in CDB
 - ▷ mu3e probably has more urgent problems in 2026
- New **global tag(s)** ready for 2026
 - ▷ **data7.1=2026V0** data from detector (**alignment?**)
 - ▷ **mcrealisticv7.1=2026V0** MC production (**alignment?**)
 - ▷ **OP naming scheme settled?**
- RDB ready
 - ▷ **rdb.py** for non-browser access
 - ▷ **fill with relevant information?! eras?**
- prompt workflow mostly ready on merlin6
 - ▷ DQM, with RDB update
 - ▷ calibration, with CDB update
 - ▷ reconstruction